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Ionic liquids as Post-Treatment Agents for Simultaneous Improvement of Seebeck Coefficient and Electrical Conductivity in PEDOT:PSS Films

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Abstract: In this study, ionic liquid (IL) post-treatment for thin films of poly(3,4-ethylene dioxothiophene): poly(styrene sulfonate) (PEDOT:PSS) is employed for the simultaneous enhancement of Seebeck coefficients and electrical conductivities. Through systematic variation of the ILs, by changing the anions while keeping the cation unchanged, changes in thermoelectric, spectroscopic and morphological properties are investigated by means of UV-Vis-spectroscopy and grazing-incidence wide-angle x-ray scattering (GIWAXS) as function of the IL concentration. The simultaneous enhancement in the two important thermoelectric properties is ascribed to the binary nature of the ILs, which complements that of PEDOT:PSS. The anions of the ILs primarily interact with the positively-charged, conducting PEDOT, while the cations interact with negatively-charged insulating PSS. Therefore, post-treatment with ILs allows for primary and secondary doping of PEDOT:PSS at the same time. Differences in the obtained Seebeck coefficients for the investigated ILs are ascribed to the chemical properties of the anions. Additionally, the choice of the latter has implications on the morphology of the treated PEDOT:PSS films regarding average π - π -stacking distances of PEDOT chains, PEDOT-to-PSS ratios and edge-on-to-face-on ratios, influencing charge transport properties macroscopically. A morphological model is presented, highlighting the influence of each IL in comparison with pristine PEDOT:PSS films.

1. Introduction

Research on conducting polymers poses both compelling challenges and promises for future applications in organic electronics. To this end, conducting polymers have been heavily investigated in recent years for their potential usage in organic and perovskite

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3 solar cells,¹⁻⁴ transparent electrodes and OLEDs⁵ and most recently also organic
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6 thermoelectrics.⁶⁻¹³ State-of-the-art thermoelectric (TE) materials mostly comprise
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9 heavier elements such as bismuth, silver, lead and different chalcogenide atoms. Usage
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12 of the typically more environment-friendly conducting polymers, which possess the
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15 potential to be produced with low cost on large scales, is especially promising with respect
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18 to application as organic TE materials for energy harvesting from waste heat. Additionally,
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21 polymers are based on carbon which makes them light-weight and potentially flexible.
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24 This is especially interesting with the prospect of wearable electronic devices and sensors
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27 based on organic materials.¹⁴⁻¹⁶
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36 Conducting polymers can be categorized e.g. according to the type of charge carriers
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38 they transport as n-type (negative charge carriers)¹⁷⁻²⁰ or p-type (positive ones)^{12,21,22}.
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42 Among the vast amount of conducting polymers, poly(3,4-ethylene
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44 dioxothiophene):poly(styrene sulfonate) (PEDOT:PSS) has attracted considerable
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47 research interest. PEDOT:PSS is most appreciated for its ability to transport positive
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50 charge carriers, and its semi-transparent properties when fabricated as thin film.²³⁻²⁵
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4 However, the most notable property of PEDOT:PSS is the potentially high electrical
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7 conductivity, which rivals that of a commonly used transparent electrode material, namely
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10 indium tin oxide (ITO). Since its discovery, researchers have applied many treatments
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13 either to the PEDOT:PSS solution prior to film deposition or post-treatments using various
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16 chemicals, ranging from high-boiling point solvents (ethylene glycol, dimethyl sulfoxide,
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19 *etc.*)^{26,27,27,27} to acids and bases^{23,25,28–32}. In most cases, the improvement in electrical
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22 conductivity originates from a multitude of factors, one of which is the decreasing content
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25 of insulating PSS posing as barrier for charge transport between the conductive PEDOT-
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28 rich domains. Furthermore, the structural order within these domains has great
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31 implications on the charge transport ability, as does the preferred orientation of the
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34 polymer chains with respect to the substrate. Charges are not only transported along the
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37 polymer backbone, but also across different chains via π - π -interactions, wherefore the
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40 knowledge of the average distances between chains and their respective predominant
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43 orientation is crucial to understanding macroscopic charge transport properties.^{33,34}
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49 Additionally, much effort has gone into the investigation of PEDOT:PSS film formation
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3 during printing³⁵, morphology evolution upon treatment with high-boiling point
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7 solvents^{33,36,37} and also the effect of humidity on pre-formed thin films³⁸.
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11 Pristine PEDOT:PSS thin films show low electrical conductivities of around 1 S cm⁻¹, and
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14 therefore are not viable for many applications. However, due to the binary nature of
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18 PEDOT:PSS, a large window of tunability of properties, such as the electrical conductivity,
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21 or the Seebeck coefficient, which is crucial for thermoelectric applications, has been
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25 opened.³⁹ For instance, an increase of the electrical conductivity up to 4000 S cm⁻¹ can
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28 be achieved by treatment of thin films with weak²⁵ and strong acids^{23,28}, solutions of
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31 inorganic salts in organic solvents²⁷, largely stemming from changes in the morphology
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35 and the PEDOT-to-PSS ratio. The electrical conductivity of a material is proportional to
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38 the charge carrier concentration, while for the Seebeck coefficient the opposite is the
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41 case. Since the PH1000 formulation of PEDOT:PSS offers potentially high electrical
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44 conductivities due to a large number of charge carriers, the obtained Seebeck coefficients
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47 are typically low. Consequently, researchers have also attempted to improve the latter
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50 quantity. As shown in our previous study, this can be done by a facile approach of mixing
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PEDOT:PSS with readily available inorganic redox-active or basic salts, leading to an effective dedoping of PEDOT and thereby increasing the Seebeck coefficient up to 70 $\mu\text{V K}^{-1}$ due to a lower number of charge carriers.³¹ Alternatively, Bubnova *et al.* used vapor treatment with tetrakis(dimethylamino) ethylene similarly, in order to tune the oxidation level of PEDOT, thereby enhancing the Seebeck coefficient.⁴⁰

Although these methods are promising in their own right and worth investigating further, they only address the improvement of one thermoelectric quantity at a time. Ionic liquids (ILs), which are ionic compounds with typical melting points below 100 °C, are mostly used as versatile solvents for chemical reactions, flame retardants and as electrolytes in lithium ion batteries.⁴¹ For our purposes, they are interesting as treating agents, since they can complement the binary nature of the PEDOT:PSS due to the former also consisting of both positively and negatively charged species. For example, Badre *et al.* combined 1-ethyl-3-methyl imidazolium tetracyanoborate (EMIM TCB) with PEDOT:PSS in solution, and were able to fabricate films with exceptional transparency properties along with a very high electrical conductivity of 2084 S cm^{-1} , rendering them promising

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3 candidates as transparent polymeric electrodes. Atomic force microscopy (AFM) and
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7 Fourier-transform infrared spectroscopy (FTIR) showed that EMIM TCB is incorporated
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10 and leads to structural rearrangement.⁴² Moreover, Murphy *et al.* performed x-ray and
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13 neutron scattering investigations on solutions of PEDOT:PSS mixed with EMIM TCB, and
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16 investigated the interactions between the binary polymer solution and the binary ionic
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19 compound.⁴³ Indeed, the loose mesh-like structure of PSS onto which PEDOT oligomers
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22 are attached is disturbed upon addition of EMIM TCB due to electrostatic interactions of
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25 the negatively charged PSS with the positively charged EMIM cation and dissociation of
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28 the ionic bond between EMIM and TCB. Therefore, implications on structure formation
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31 and final film morphology are to be expected when using ILs. This notion is supported by
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34 the work of de Izarra *et al.*, who used DFT calculations to predict interactions of
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37 PEDOT:PSS with different EMIM-based ILs, which confirmed the findings made by Kee
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45 *et al.*^{44,45}
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49 In our present study we not only investigate the effect of IL addition on the morphology of
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52 PEDOT:PSS thin films, but also on the respective thermoelectric properties. Since it has
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4 been shown that the electrical conductivities of PEDOT:PSS films increase upon
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7 treatment with ILs, it will be interesting to see if there is an interaction between PEDOT,
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10 which is mostly responsible for the thermoelectric properties, especially the Seebeck
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13 coefficient, with the anionic component of the ILs. Additionally, we chose three different
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17 ILs on a rational basis, in which the cation is always EMIM, and only the anion is changed.
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21 Pre-fabricated PEDOT:PSS thin films are post-treated with solutions of the ILs 1-ethyl-3-
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24 methyl imidazolium dicyanamide (EMIM DCA), 1-ethyl-3-methyl imidazolium
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27 tetracyanoborate (EMIM TCB) and 1-ethyl-3-methyl imidazolium tetrafluoroborate (EMIM
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30 BF_4) in tetrahydrofuran (THF) as solvent. In many studies in which ILs were used for
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34 treatment of PEDOT:PSS, the samples were extensively washed with water in order to
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38 remove remnants of the former. ILs typically possess a large temperature range in which
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42 they are stable in liquid phase⁴¹, which is why they cannot be evaporated from the films
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46 using thermal annealing procedures. However, since the treatment of PEDOT:PSS with
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49 water changes the films' properties, we opted for THF as solvent to prove its viability as
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53 a carrier of ILs for post-treatment. To this end, PEDOT:PSS thin films are post-treated
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57 with pure THF as reference sample in addition to solutions of the three above-mentioned
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3 ILs in different concentrations with THF as solvent. Post-treatment is chosen over direct
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7 mixing of PEDOT:PSS and the ILs, due to the expected interactions of the components
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10 and therefore possible impediment of film fabrication. The post-treated films are then
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13 subjected to Seebeck coefficient and electrical conductivity measurements, before a
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16 washing step with pure THF is performed. This washing step ensures complete removal
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19 of excess ILs from the films, in order to prove that changes in the thermoelectric
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22 capabilities of the films stem from permanent changes of the polymer films and not from
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25 the presence of residual ILs on the surfaces. This will be seen later on, in the discussion
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28 of structural changes. After that, the samples are again subjected to a thermoelectric
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31 characterization, in addition to UV-Vis spectroscopy and grazing-incidence wide-angle x-
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34 ray scattering (GIWAXS) to investigate changes in the structure. Finally, a morphological
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37 model is proposed in an attempt to link the changes in the thermoelectric properties to
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40 the choice of IL, and more specifically, if the choice of anion only influences PEDOT:PSS
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43 on an electronic level or also on a morphological one.
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2. Thermoelectric characterization

Figure 1 shows a sketch of the sample preparation process. PEDOT:PSS solutions were prepared and spin-cast on clean soda-lime glass substrates, and subjected to thermal annealing afterwards, as reported previously^{24,31,46}. This process was repeated in order to gain a sufficient film thickness of about 150 nm. Post-treatment of the thin films was performed by preparing dilutions of the three ILs with 1-ethyl-3-methyl imidazolium (EMIM) as cation, together with the anions dicyanamide (DCA, green), tetracyanoborate (TCB, orange) and tetrafluoroborate (BF₄, blue) in tetrahydrofuran (THF) as solvent. The solutions were drop-cast on pre-fabricated PEDOT:PSS thin films, left to take effect for one minute, and then spun-off. Next, the samples were annealed at elevated temperatures of 140 °C, in order to remove residual THF. After measuring the Seebeck coefficient and electrical conductivity, samples were washed with pure THF to rule out that the observed changes in the thermoelectric properties originate from the presence of residual ILs on the sample surfaces. Due to the typically very high boiling points of ILs, one cannot expect them to evaporate at the chosen annealing temperature of 140° C. UV-Vis spectroscopy and GIWAXS were performed after this last THF washing step, and the thermoelectric characterization was repeated.

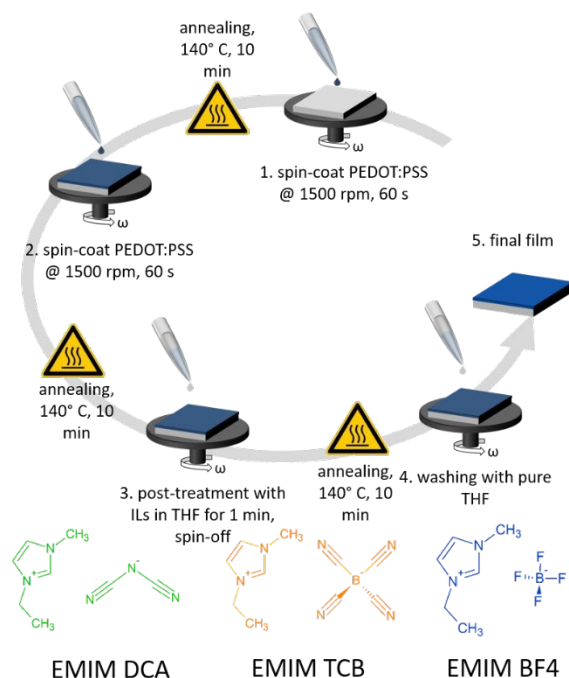


Figure 1. Sketch of preparation of PEDOT:PSS thin films post-treated with solutions of three ionic liquids dissolved in THF. This step is followed by washing with pure THF, in order to remove any remaining ionic liquids from the film surface. The color-coding represents the different ionic liquids for the remainder of this work.

In addition, a reference PEDOT:PSS film was prepared by performing the post-treatment step with pure THF, in order to verify that the observed changes are not related to interactions of the solvent with the PEDOT:PSS film.

Due to the binary nature of both, the ionic liquids and PEDOT:PSS, the former were chosen rationally. One can expect the positively charged EMIM cations of the ionic liquids to interact primarily with the negatively charged PSS through electrostatic interactions, while the anions are expected to primarily interact with PEDOT, with the latter providing the macroscopic electrical conductivity by carrying positive charge carriers. In this work, we want to investigate the influence of the anions on the thermoelectric properties of treated PEDOT:PSS thin films, since their interaction with the conducting PEDOT should have ramifications on the Seebeck coefficient and

electrical conductivity due to possible changes in charge carrier concentration, as a result of primary doping effects.

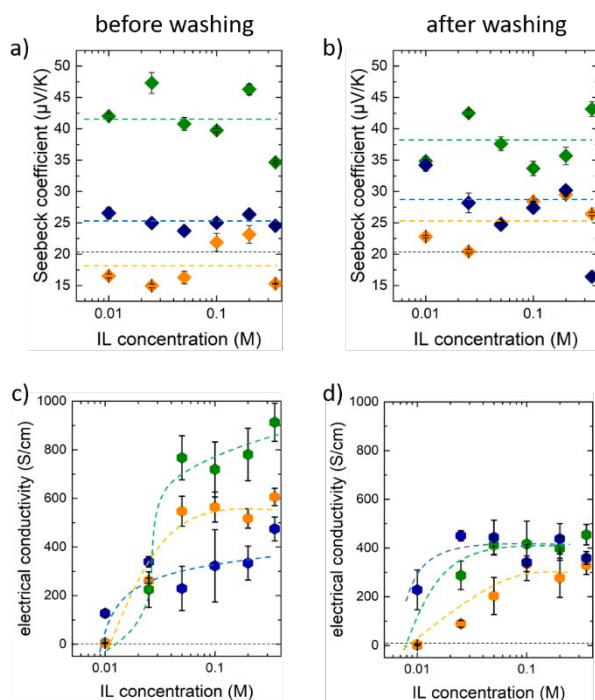


Figure 2. Thermoelectric properties of PEDOT:PSS thin films post-treated with solutions of ionic liquids in THF in different concentrations: Seebeck coefficients a) before and b) after washing with pure THF, and electrical conductivities c) before and d) after washing with pure THF. Data are shown for PEDOT:PSS films treated with EMIM DCA (green), EMIM TCB (orange) and EMIM BF₄ (blue), respectively. Dashed black lines indicate reference values of PEDOT:PSS treated with pure THF, while colored, dashed lines serve as guides to the eye.

Figure 2a shows the Seebeck coefficient for treated PEDOT:PSS films before the washing step.

While there seems to be no clear concentration dependence of the Seebeck coefficient on either of the ILs, one can easily spot distinct average levels for the different ILs, which are each highlighted by guides to the eye. The values for EMIM DCA-treated samples are around $42 \mu\text{V K}^{-1}$, for EMIM BF₄-treated ones around $25 \mu\text{V K}^{-1}$, and for EMIM TCB-treated ones around $18 \mu\text{V K}^{-1}$. As shown in Figure 2b, the washing step changes the measured Seebeck coefficients. Average values for EMIM DCA-treated samples now are around $38 \mu\text{V K}^{-1}$, EMIM BF₄-treated ones around $27 \mu\text{V K}^{-1}$ and EMIM TCB-treated ones around $25 \mu\text{V K}^{-1}$. Thus, after the washing step with pure THF

the Seebeck coefficients are improved as compared to the reference value of $21 \mu\text{V K}^{-1}$, which proves that the properties of PEDOT:PSS are permanently altered due to the treatment with ionic liquids. A limited number of sites at which the anions attach to PEDOT chains could explain the absence of a concentration dependence.

In their recent work, Fan *et al.* post-treated PEDOT:PSS thin films with sulfuric acid, sodium hydroxide and different ILs dissolved in methanol.⁴⁷ They observed increases in both the Seebeck coefficient and electrical conductivity, leading to high power factors. Their values for the Seebeck coefficients after treatment with EMIM DCA and EMIM BF_4 closely match our values, with a strong difference being their observed electrical conductivities. This stems from their preceding treatment with sulfuric acid, which is known to strongly alter the morphology of PEDOT:PSS and subsequently lead to high electrical conductivities. Additionally, methanol affects PEDOT:PSS in a similar, although not as drastic way. For these reasons, we chose THF as solvent for the ILs, since it alters neither the Seebeck coefficient nor the electrical conductivity by itself. We do not believe the enhancement of the Seebeck coefficient to originate from ion accumulation due to the temperature gradient, as the enhancement is still observed after the washing step with pure THF. The contribution of the ionic Seebeck effect, closely studied by Wang *et al.*,⁴⁸ is expected not to play any significant role, as was also concluded by Fan *et al.*⁴⁷ No time-dependence of the magnitude of the Seebeck coefficient was observed during the measurement, which proceeded for each sample for at least 20 minutes. Ionic Seebeck effects are typically found in the mV K^{-1} range, whereby values of this magnitude cannot be found here.

In Figure 2c, an increase in the electrical conductivity upon treatment with all ionic liquids before the washing step is apparent. In general, one observes an increase in conductivity with increasing

IL concentration resulting in higher values as compared to the pristine PEDOT:PSS sample, which has an electrical conductivity below 1 S cm^{-1} . For EMIM DCA, an increase from 5 S cm^{-1} for low concentrations up to 913 S cm^{-1} for high concentrations is observed. The maximum conductivity that can be achieved by treatment with EMIM TCB is around 600 S cm^{-1} . In contrast, Badre *et al.* achieved semi-transparent thin films with electrical conductivities larger than 2300 S cm^{-1} , when they mixed PEDOT:PSS solutions directly with EMIM TCB.⁴² Moreover, they used significantly larger concentrations above 30 wt% as compared to the present study. A comparatively high electrical conductivity of approximately 130 S cm^{-1} is obtained for low concentrations of EMIM BF_4 , which, however, levels off at 480 S cm^{-1} for high concentrations. In order to demonstrate the lasting changes made on the PEDOT:PSS films upon IL post-treatment, the samples are again measured after washing with pure THF, as shown in Figure 2d. A decrease in the maximum conductivities of PEDOT:PSS films treated with EMIM DCA and EMIM TCB is found, leading to values of approximately 450 S cm^{-1} in the former and 330 S cm^{-1} in the latter case upon washing. Interestingly, the conductivities for samples treated with EMIM BF_4 are even slightly improved by this washing process. Again, the overall improved conductivities of the PEDOT:PSS films even after washing prove that permanent changes in their properties have been installed by treatment with ILs. The low intrinsic conductivities of the selected ILs (see Table S1) cannot explain the conductivity values observed for treated PEDOT:PSS films. By making use of the binary nature of both, PEDOT:PSS and the different ILs, we are able to simultaneously increase the Seebeck coefficients and the electrical conductivities of the treated polymer films. Careful assessment of graphs in which the electrical conductivity and Seebeck coefficient are shown against each other do not indicate a direct correlation. This is in direct contrast to our previous work, in which inorganic salts were used to alter the charge carrier concentration in PEDOT:PSS thin films,

yielding an inverse behavior of Seebeck coefficient and electrical conductivity.³¹ Thus, the findings of the present work are counterintuitive, since one typically expects an inverse correlation between Seebeck coefficient and electrical conductivity. A combination of so-called primary doping (altering of charge carrier concentration) and secondary doping (altering of film morphology and improvement in charge carrier mobility) is suggested as possible mechanism for our observations.

Figure 3 shows the power factor (σS^2) values for treatment with each IL, before (solid symbols) and after (hollow symbols) the washing with pure THF. An alternative representation is given in **Figure S1**. A maximum power factor of $167 \mu\text{W K}^{-2} \text{m}^{-1}$ is found for high concentrations of EMIM DCA, as shown in Figure 3a, which stems from the simultaneous strong increase in Seebeck coefficients and electrical conductivities. The washing step decreases the Seebeck coefficient slightly and the electrical conductivity strongly, which manifests itself in overall decreased power factors, of up to $85 \mu\text{W K}^{-2} \text{m}^{-1}$ for the highest concentration of EMIM DCA.

The power factors of PEDOT:PSS films treated with EMIM TCB are considerably lower, as can be seen from Figure 3b. The maximum power factors before and after washing are around $27 \mu\text{W K}^{-2} \text{m}^{-1}$ for intermediate concentrations, in both cases. Even though electrical conductivities decrease after washing, a simultaneous increase of the Seebeck coefficient is observed. The improvement compared to pristine PEDOT:PSS stems from an increase in electrical conductivity, while maintaining similar Seebeck coefficients.

Lastly, power factors for EMIM BF_4 -treated samples (shown in Figure 3c), are mostly constant, both before and after the washing. In the former case, the maximum power factor is $29 \mu\text{W K}^{-2} \text{m}^{-1}$, while in the latter case the values are increased overall, giving a maximum of $40 \mu\text{W K}^{-2} \text{m}^{-1}$. The increase in power factor values upon washing can be explained by the slightly increased

Seebeck coefficients and additionally the improved electrical conductivities. The highest power factor is found for the second-highest concentration of 0.2 M EMIM BF₄ in THF, since for the highest concentration the Seebeck coefficient is considerably decreased after washing.

Summarizing, one can say that post-treatment of PEDOT:PSS films with ILs is a promising approach, while the strong impact of IL choice is also apparent. The type of anion not only influences the Seebeck coefficient, but also the electrical conductivity. In all cases, an improvement of the power factor compared to the pristine sample is observed.

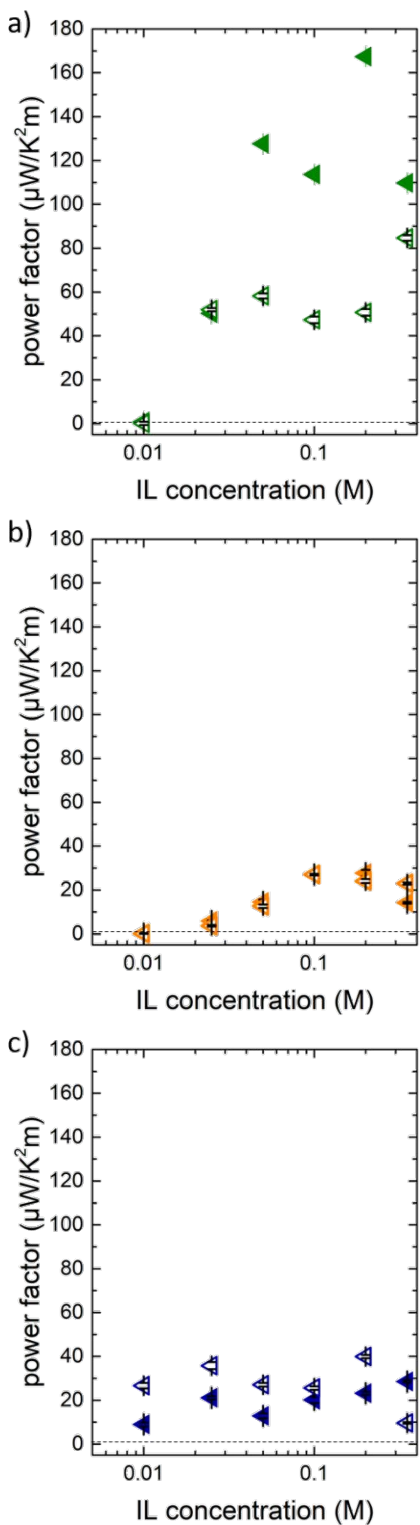


Figure 3. Power factors of PEDOT:PSS thin films post-treated with solutions of a) EMIM DCA, b) EMIM TCB and c) EMIM BF₄ in different concentrations in THF. Solid symbols represent the values before, hollow symbols represent the values after washing with pure THF. Black lines indicate reference values of PEDOT:PSS treated with pure THF, respectively.

3. Spectroscopic characterization

After making the general observation of changes upon treatment of PEDOT:PSS thin films with ILs, a deeper look into the improvement mechanisms is required. As discussed by Kang *et al.*, the Seebeck coefficient is inversely related to the number of charge carriers in the sample.⁴⁹ Changes in the former suggest altering of charge carrier density upon IL post-treatment of PEDOT:PSS films.

The PH1000 formulation of PEDOT:PSS possesses a PEDOT-to-PSS ratio such that maximum conductivity can be achieved with appropriate treatments. This leads to the typically low Seebeck coefficients one acquires with these kinds of samples. Due to the strong doping of PEDOT chains with PSS, the charge carrier density is very high, or in other words, bipolarons are the dominant charge carrier type.^{31,50} Bipolarons represent the second oxidation state of PEDOT, and carry a double positive charge. A high bipolaron concentration is typically accompanied by strong optical absorption in the infrared-range. The smaller the charge carrier concentration becomes, the more the absorption shifts towards the visible range.^{31,50} UV-Vis spectroscopy has often been used to gain insights into the different types of charge carriers and their relative amounts in PEDOT:PSS. As seen in our previous study of PEDOT:PSS films dedoped using inorganic salts, new absorption features appeared in the UV-Vis spectra in the near infrared (around 950 nm) and in the visible range (around 600 nm).³¹ These absorption features were assigned to the presence of polaronic and neutral states, respectively. We linked this to a gradual loss of charge carriers upon increasing dedopant concentration, which coincided with increasing Seebeck coefficients in all cases.³¹

Since we expect the anionic component of the ILs to primarily interact with PEDOT chains and alter the charge carrier density, we perform UV-Vis spectroscopy as a facile method for qualitatively determining the doping level in IL post-treated PEDOT:PSS films. The

corresponding results for the three different ILs are shown in **Figure 4**, with an alternative representation given in **Figure S2**.

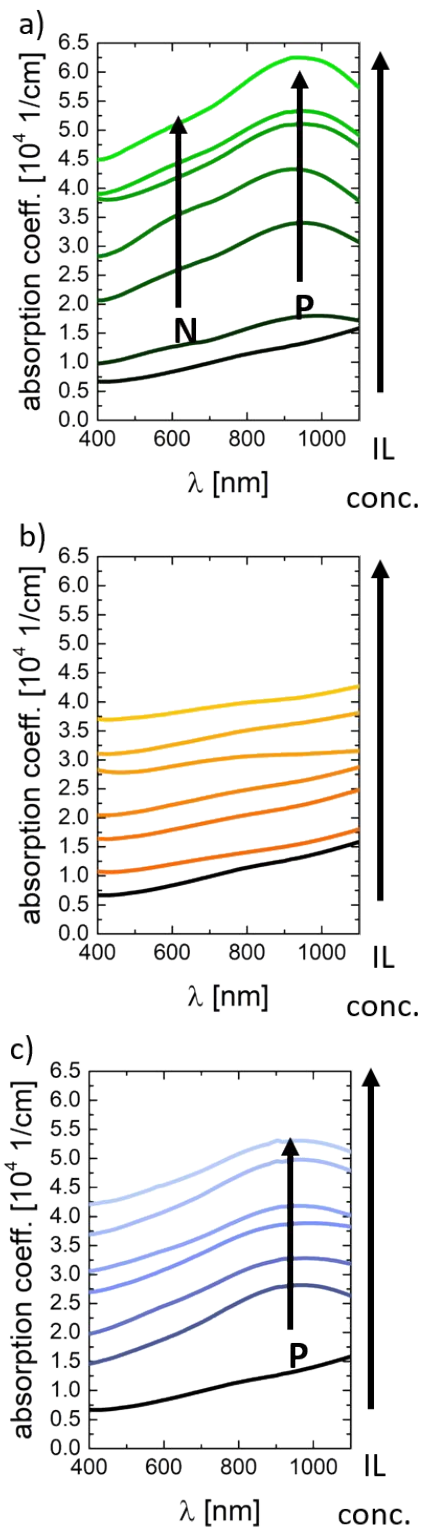


Figure 4. UV-Vis spectra of PEDOT:PSS thin films treated with a) EMIM DCA, b) EMIM TCB and c) EMIM BF₄, after washing with THF. The curves are shifted vertically for clarity, with the respective increasing concentrations of ionic liquids also visualized by gradually brighter colors. Black curves represent the reference spectrum of PEDOT:PSS not treated with ionic liquids. Arrows marked with P and N represent absorption features originating from polaronic and neutral states, respectively.

In Figure 4a, the UV-Vis spectra for EMIM DCA-treated samples are shown, along with the reference curve for pristine, untreated PEDOT:PSS. The shoulder on the infrared side of the spectrum of the reference hints at the dominance of bipolarons as charge carriers, as discussed above. However, even upon treatment with small concentrations of EMIM DCA (0.01 M), strong changes of the spectra become observable. Treatment with EMIM DCA transforms the shoulder extending into the infrared range into an absorption peak centered around 950 nm, which can be assigned to polaronic states (as marked by the P-arrow in Figure 4a). Additionally, features centered around 600 nm are assigned to neutral states (as marked by the N-arrow in Figure 4a). The drastic change of the spectra even upon treatment with small concentrations of EMIM DCA is in line with the prompt increase of the Seebeck coefficient, also observed for small concentrations.

In contrast to this, EMIM TCB-treated PEDOT:PSS films exhibit no changes in their absorption behavior as function of EMIM TCB concentration, as can be seen in Figure 4b. The shape of the reference curve is retained, which hints at an absence of changes in the charge carrier distribution. Interestingly, again a different behavior is found for EMIM BF₄-treated PEDOT:PSS samples. While a similar absorption feature emerges at approximately 950 nm with low concentrations of IL, and is retained with higher concentrations, indicating the presence of polaronic species in all cases. The feature centered around 600 nm is much weaker than for the samples treated with EMIM

DCA. This leads to the conclusion that while the treatment with EMIM BF₄ leads to the formation of polaronic species, neutral species are formed only to small degree, if at all.

In general, the discussion of the UV-Vis spectra allows for a link to the observed changes in the Seebeck coefficients of the differently treated PEDOT:PSS films. For our system, no clear dependence of the Seebeck coefficients on IL concentrations is found, despite the fact that the type of IL affects the average Seebeck coefficients. The observed Seebeck coefficients are highest for samples treated with EMIM DCA, followed by the ones treated with EMIM BF₄ and lastly the ones treated with EMIM TCB. Looking at the UV-Vis spectra this coincides with the presence of both polaronic and neutral species in the case of EMIM DCA, thus indicating low relative numbers of charge carriers. Following, the spectra of EMIM BF₄-treated samples primarily exhibit absorption features linked to polaronic species. In line with this, EMIM TCB-treated samples, which show the lowest average Seebeck coefficients, do not show significant changes of the spectra as a function of EMIM TCB concentration.

A possible explanation in these trends might be given by molecular properties of the used anions. The DCA anion shows an angulated structure, with its negative charge located at the central nitrogen atom. In contrast, both the BF₄- and TCB-anion possess tetrahedral symmetry, with the negative charge being located at the center. In addition, the biatomic nature of the cyano-ligands in the TCB-anion leads to a much larger size, and thus stronger delocalization of the charge, compared to the BF₄-anion. According to the systematic study, which was performed by Cremer *et al.*, the structure of the anion has implications on the properties of the ILs. In this case, the DCA-anion (in which the negative charge is strongly localized on the central nitrogen atom) shows the highest basicity and thus the strongest interaction with PEDOT, followed by the BF₄-anion (intermediate charge localization) and the TCB-anion (weak charge localization, due to larger

size).⁵¹ Additionally, the systematic study of Spange *et al.* gives credence to the above-mentioned hypothesis.⁵² The hydrogen-bond basicity (i.e. the ability of the anions to use their negative charge for interactions with cations) is highest for DCA, lower for BF₄ and lowest for the TCB anion. In this order, their ability to increase the electron density on PEDOT chains decreases. This is linked the reduction in the average oxidation level, which directly affects the Seebeck coefficient. Thus, a qualitative link between UV-Vis spectra and the Seebeck coefficients of the corresponding samples is found, with deviations from the bipolaronic nature of PEDOT:PSS being linked to a decrease in charge carrier concentration and subsequent increase in Seebeck coefficients.

4. Morphological characterization

Apart from the electronic properties of IL-treated PEDOT:PSS films, which primarily explain changes in the Seebeck coefficient, it has been shown many times that changes in the film structure upon treatment are responsible for the generally found increase of electrical conductivities compared to pristine PEDOT:PSS. This led us to perform GIWAXS measurements on IL-treated PEDOT:PSS films in order to gain insights into the mechanism of conductivity enhancement, and also into possible differences between the different ILs. In contrast to surface sensitive methods such as x-ray photoelectron spectroscopy, in GIWAXS the x-rays penetrate the entire film, because an incident angle above the critical angle was chosen. Moreover, the elongated footprint of the x-ray beam in GIWAXS geometry enables to sample a large sample volume. Thus, the information obtained from the GIWAXS measurements possesses a high statistical relevance.^{53–56}

PEDOT:PSS is known to exhibit anisotropic conduction properties, which mostly stems from the two-fold way, PEDOT stacks arrange in a thin film. PEDOT chains typically form supermolecular aggregates via π - π -interactions, whereas the two distinct orientations that are found are named

face-on or edge-on, indicating the predominant direction for inter-chain charge transport, respectively.^{34,35} Correspondingly, 2D GIWAXS images can be analyzed in vertical direction, which reveals information about the crystallinity of PEDOT oriented in face-on direction, and in horizontal direction, revealing information about the crystallinity of PEDOT in edge-on direction. The main quantities, which are of interest for this study and can be extracted from the GIWAXS data, are π - π -stacking distances, the PEDOT-to-PSS ratio, and the ratio of crystallites oriented in edge-on and face-on directions.

The π - π -stacking distance relates to the orbital overlap between chains, and thereby improves the inter-chain charge transport the smaller it becomes. A larger PEDOT-to-PSS ratio in general also leads to improved electrical conductivities, since a larger ratio hints at more efficient removal of the insulating PSS, yielding better interconnection between conducting domains. Lastly, since all thermoelectric quantities are measured in an in-plane measurement geometry, a large edge-on-to-face-on ratio is also correlated to improved electrical conductivities, since inter-chain transport along the sample plane becomes more efficient, the more PEDOT stacks are oriented in edge-on direction.

4.1. π - π -stacking of PEDOT chains

All quantities are extracted from cuts in the 2D GIWAXS data made along the q_z -direction for analysis of face-on crystallites, and along the q_r -direction for analysis of edge-on crystallites, which are shown in **Figure S4** and **S5**, respectively. The curves for the IL-treated samples are shown together with the pristine PEDOT:PSS reference. No scattering signals can be assigned to structural correlation between anions and cations of the ILs, from which the complete removal of the latter with the THF washing step is concluded. For all three ionic liquids, scattering features at

low q -values in face-on direction become apparent with increasing concentration (marked with arrows in Figure S4), hinting at improved order in the vertical layering.^{35,57} In the case of EMIM DCA, this peak is centered around $0.45 \text{ \AA}^{-1}$, corresponding to a characteristic distance of approximately $14 \text{ \AA}$, with this peak already arising for the lowest concentration. Similarly, a peak centered at $0.41 \text{ \AA}^{-1}$ is also found for EMIM TCB-treated samples, however, only for intermediate concentrations. This position corresponds to a slightly larger characteristic distance of approximately $15 \text{ \AA}$, which increases up to $18 \text{ \AA}$ for high concentrations of EMIM TCB. Although less pronounced, a peak can also be discerned for EMIM BF_4 -treated samples, which is centered around $0.31 \text{ \AA}^{-1}$, already for low concentrations. Corresponding to approximately $20 \text{ \AA}$, it exhibits the largest characteristic distance for the vertical layering. Interestingly, only for EMIM TCB- and EMIM BF_4 -treated samples higher order peaks at doubled peak position values are found, which hints at extended order of vertical layering for these treatments.

Figure 5 shows the extracted values for π - π -stacking distances extracted from closer analysis of the curves in the high- q region for both, face-on and edge-on directions. PEDOT chains are known to form super-molecular aggregates via π - π -interactions, which determine the efficiency of charge transport between different chains.³³ Typically, the denser the packing of chains in these aggregates becomes, the more efficiently charges are transported due to

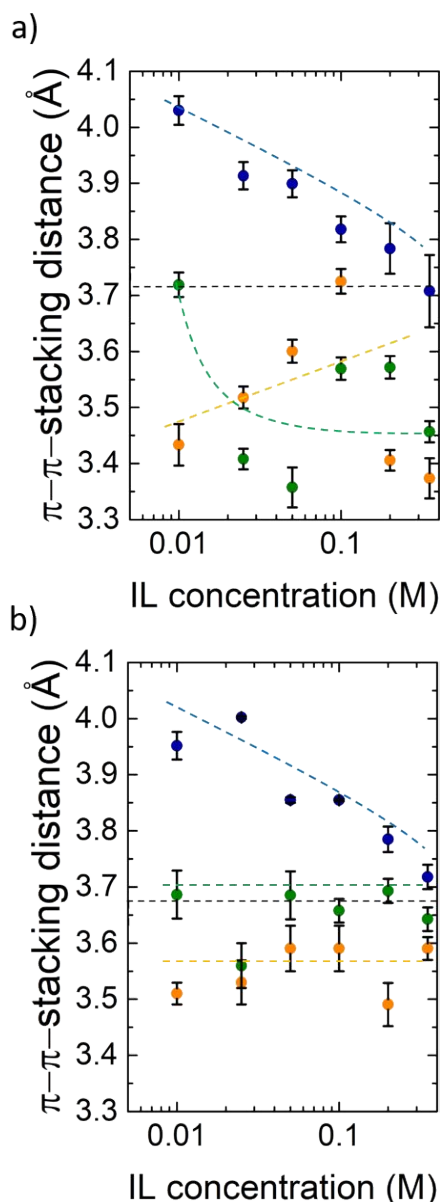


Figure 5. π - π -stacking distances obtained from GIWAXS measurements in a) face-on direction and b) edge-on direction for samples treated with the three ionic liquids EMIM DCA (green), EMIM TCB (orange) and EMIM BF4 (blue) in THF in different concentrations. Dashed black lines represent the reference values for PEDOT:PSS not treated with ionic liquids.

increased overlap of π -orbitals.³³ For this study, it is of interest to investigate possible changes in the π - π -stacking distances as function of both type and concentration of ILs. In Figure 5a, π - π -stacking distances in face-on direction are shown, with the reference for pristine PEDOT:PSS marked with the dashed black line. It becomes apparent that the π - π -stacking distances for EMIM

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3 BF_4 -treated samples are much larger than that of the reference sample and for the other ILs. With
4 increasing concentration of EMIM BF_4 , the π - π -stacking becomes increasingly dense, with
5 distances decreasing from approximately 4 Å to 3.7 Å, which is close to the reference value. EMIM
6 DCA-treated PEDOT:PSS films show much smaller stacking distances of down to 3.4 Å,
7 decreasing as function of concentration. While there is a weaker concentration dependence for
8 EMIM TCB-treated PEDOT:PSS films, the π - π -stacking distances are smaller in most cases,
9 compared to the reference value. Thus, in face-on direction inter chain transport is improved for
10 EMIM DCA- and EMIM TCB-treated samples as compared to the reference.

11
12 Figure 5b shows the π - π -stacking distances in edge-on direction. Here, the trend upon treatment
13 with EMIM BF_4 appears very similar to the face-on case, with π - π -stacking distances decreasing
14 from 4 Å to 3.7 Å as function of concentration. The π - π -stacking distances after treatment with
15 EMIM DCA are quite close to that of pristine PEDOT:PSS, which is 3.7 Å. Again PEDOT:PSS
16 treated with EMIM TCB shows a very dense packing, since the π - π -stacking distances for those
17 samples are centered around 3.55 Å. In edge-on direction, this again leads to the improvement of
18 the inter-chain transport for EMIM DCA and EMIM TCB.

4.2. PEDOT-to-PSS ratio

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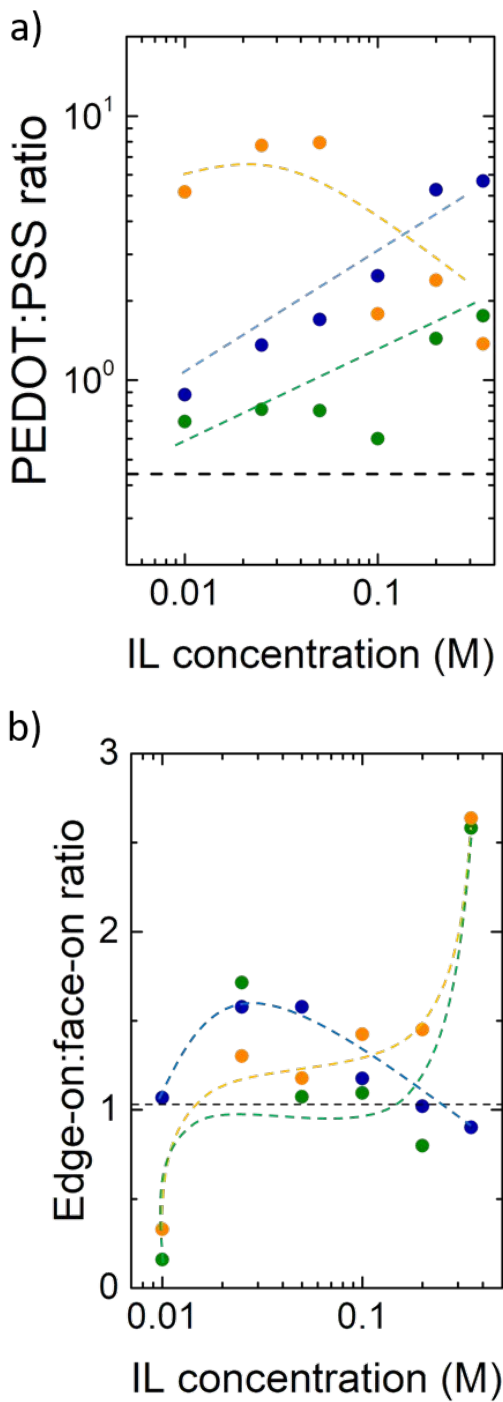


Figure 6. a) PEDOT-to-PSS ratio obtained from powder-like azimuthal integration of intensity and b) edge-on-to-face-on ratio obtained from GIWAXS for PEDOT:PSS films treated with EMIM DCA (green), EMIM TCB (orange) and EMIM BF4 (blue) in THF. Dashed black lines represent reference values obtained from pristine PEDOT:PSS films, respectively.

GIWAXS allows for the analysis of the relative amounts of PEDOT and PSS through comparison of Bragg peak intensities, assuming a constant crystalline fraction within the film. Both components exhibit scattering signals which correspond to their local order. The ratio between PEDOT and PSS is expected to have implications on the charge transport efficiency, since charge carriers cannot be transported along the electrically insulating PSS chains. Anionic PSS is expected to primarily interact with the EMIM cations of the ILs, and thereby be removed from the film more or less efficiently during sample preparation. Generally speaking, a higher PEDOT-to-PSS ratio would be favorable for an improved charge carrier transport, respectively higher electrical conductivity.

In order to obtain a full picture powder-like azimuthal integration was performed on the 2D GIWAXS patterns. Using the obtained scattering curves in Figure S4 and S5, upon analysis of the characteristic signals of PEDOT and PSS, changes in their ratio can be discerned, as seen in **Figure 6a**. The reference value for the pristine PEDOT:PSS film of approximately 0.4 matches very well with the nominal value for the PEDOT-to-PSS ratio of the polymer blend solution. Compared to that, all ILs exhibit an improvement of the PEDOT-to-PSS ratio upon post-treatment of PEDOT:PSS films. Thus, removal of PSS through its electrostatic interaction with the positively charged EMIM ions is suggested. More specifically, EMIM DCA shows an increasing trend of the ratio with concentration, whereby the increase is even more pronounced for EMIM BF₄-treated samples. The PEDOT-to-PSS ratio of EMIM TCB-treated samples on the other hand shows a slightly decreasing trend with IL concentration, indicating a less efficient removal of PSS.

In conclusion, the choice of the anion does not only influence the final films on an electronic level, but also the former's ability to extract PSS. It can also be concluded that the ability of the EMIM cation in all three ILs to extract PSS from the thermoelectric thin film possesses a comparatively

small dependence on the type of anion contained in the IL. However, since an improvement over the reference is seen in all cases, it can be assumed that the increase in electrical conductivity is aided by the increasing PEDOT-to-PSS ratio due to loss of insulating material.

4.3. Changes in molecular orientation of PEDOT

The last quantity of interest is the ratio between PEDOT chains oriented in edge-on direction to PEDOT chains oriented in face-on direction (**Figure 6b**). Scattering intensities of the characteristic PEDOT signal are compared in edge-on and face-on direction. In pristine, untreated PEDOT:PSS there is a balance between PEDOT oriented in both face-on and edge-on direction. However, this changes upon post-treatment with ILs. EMIM DCA- and EMIM TCB-treated PEDOT:PSS films show similar trends as function of concentration, with the face-on orientation being prevalent for low concentrations, slightly increasing the relative amount of PEDOT oriented in edge-on direction compared to the reference, while primarily exhibiting an edge-on orientation for high IL concentrations. For post-treatment with EMIM BF₄, only for intermediate concentrations the edge-on-to-face-on ratio is improved, otherwise being close to the reference value. Summarizing, the edge-on-to-face-on ratio is improved for high concentrations of EMIM DCA and EMIM TCB and for intermediate concentrations of EMIM BF₄.

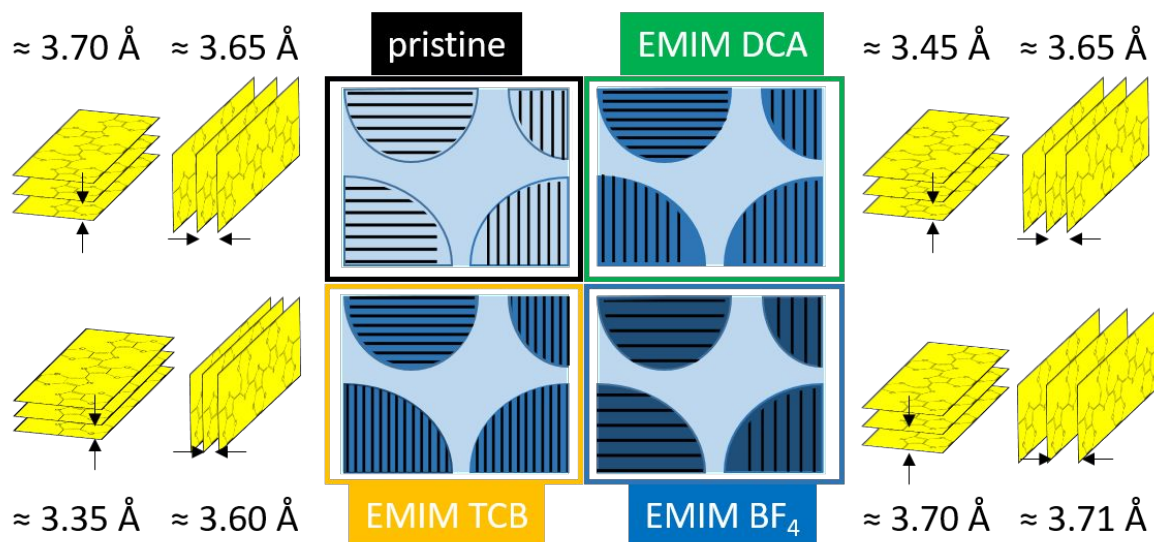


Figure 7. Structural model of PEDOT:PSS (PEDOT: dark blue, PSS: light blue) in its pristine state (top left), and treated with the highest concentrations each of EMIM DCA (top right), EMIM TCB (bottom left) and EMIM BF₄ (bottom right). Schematic stacks of PEDOT on each side of the sketch show π - π -stacking distances in face-on and edge-on direction, respectively. Black lamellae indicate direction of predominant stacking of PEDOT chains via π - π -stacking. The colors of the domains represent the PEDOT-to-PSS ratio of the domains, with darker domains representing a higher PEDOT-to-PSS ratio. The edge-on-to-face-on ratio is reflected in the ratio of domains exhibiting stacking in edge-on and face-on direction.

4.4. Morphological model: morphology-function relation

In order to explain the different trends observed for the electrical conductivities of IL-treated PEDOT:PSS films, one has to take into account the complex interplay of different factors. **Figure 7** aims to visually merge the above discussed findings, by comparing the pristine PEDOT:PSS to the IL-treated ones in terms of π - π -stacking distances, PEDOT-to-PSS ratios and edge-on-to-face-on ratios. It should be noted that the morphological model gives a schematic overview over the thin films treated with the highest concentrations of each IL. This is done in order to alleviate some of the complexity of the dataset, and highlight the differing influences of each individual IL. Pristine PEDOT:PSS (top left part of Figure 7) possesses a balance of PEDOT oriented in both face-on and edge-on direction, as indicated by the black lamellae stacked in these orientations.

Since the pristine PEDOT:PSS film is not post-treated, the PEDOT-to-PSS ratio is very small, which is indicated by the light blue color of the domains representing more PSS. As will be discussed later, with π - π -stacking distances around 3.70 Å and 3.65 Å in face-on and edge-on direction, respectively, pristine PEDOT:PSS lies in the intermediate range of possible stacking distances.

Treatment with EMIM DCA leads to a surplus of PEDOT oriented in edge-on direction, while the π - π -stacking distance is very similar to that of pristine PEDOT:PSS. PSS is efficiently removed by EMIM DCA, which is reflected in the higher PEDOT-to-PSS ratio. In face-on direction, the π - π -stacking distances of around 3.45 Å are much smaller than for the reference. Therefore, the increasing electrical conductivities most likely originate from a combination of having more PEDOT oriented in edge-on direction especially for high EMIM DCA concentrations, in addition to having less insulating PSS.

EMIM TCB-treated PEDOT:PSS films show denser π - π -stacking in face-on orientation and slightly denser stacking in edge-on direction. In addition, here a re-orientation towards more edge-on orientation can be observed. The PEDOT-to-PSS ratios, while being high for small concentrations, decrease as function of concentration. This suggests that insulating PSS is less efficiently removed with higher concentrations of EMIM TCB, and might explain why the electrical conductivity, while higher compared to the reference, is lower than for the other ILs.

Treatment with low concentrations of EMIM BF₄ leads to the largest π - π -stacking distances of approximately 4.0 Å in this study, for both face-on and edge-on orientation. These distances reach the values of pristine PEDOT:PSS with high concentrations of EMIM BF₄. The edge-on-to-face-on ratio is close to the reference value for low concentrations and is therefore much higher than for the other ILs in this range. For intermediate concentrations, the edge-on orientation even seems

to be favored. The PEDOT-to-PSS ratio increases with EMIM BF₄ concentration, while the obtained values are larger than those for EMIM DCA. The comparatively high electrical conductivity for low concentrations of EMIM BF₄ can be explained by a combination of the decent edge-on-to-face-on ratio and high PEDOT-to-PSS ratio. For high concentrations, the overall PEDOT-to-PSS ratio is greatly improved, while again having a decently high edge-on-to-face-on ratio. A possible explanation for the lower saturation conductivities in the EMIM TCB case might be less efficient removal of PSS, since both π - π -stacking distances as well as the edge-on-to-face-on ratios change favorably as function of EMIM TCB concentration.

Additionally, lamellar stacking, whose corresponding signals are found at small q-values in the scattering curves, is also expected to influence the electrical conductivity. EMIM DCA, EMIM TCB and EMIM BF₄ show increasing lamellar stacking distances in this order. Similarly to charge transport in π - π -stacks, smaller distances of lamellae can lead to an overall improved electrical conductivity. In general, there is a complex interplay between the different structural parameters and their respective effect on the macroscopically measurable quantities. Non-linear dependencies are expected, making it challenging to quantitatively gauge the influence of each parameter. Instead, this study focuses more on the qualitative link between structure and electrical conductivity. In summary, treatment with all three ILs improves the structure of the respective PEDOT:PSS films in a way that charge transport is facilitated through the samples.

5. Conclusion

Our study aimed at a systematic variation of anions in ILs for enhancement of the thermoelectric properties of PEDOT:PSS through a post-treatment procedure. Due to the binary nature of both PEDOT:PSS and the ionic liquids, simultaneous improvement of Seebeck coefficients and

electrical conductivities have been expected. For this purpose, EMIM DCA, EMIM TCB and EMIM BF₄ have been dissolved in THF in different concentrations and used for treatment of PEDOT:PSS films in order to investigate their influence on thermoelectric, spectroscopic and morphological properties of the polymer films. The highest power factors of 82 $\mu\text{W K}^{-2}\text{m}^{-1}$ are obtained for treatment with EMIM DCA (roughly 170 $\mu\text{W K}^{-2}\text{m}^{-1}$ before the washing step with pure THF), due to a simultaneous increase of Seebeck coefficient up to approximately 42 $\mu\text{V K}^{-1}$ and electrical conductivity up to 420 S cm^{-1} after washing. Power factors of up to 40 $\mu\text{W K}^{-2}\text{m}^{-1}$ are obtained for treatment with EMIM BF₄, again due to simultaneous increases in Seebeck coefficient and electrical conductivity, although to a lesser extent. In both cases, UV-Vis spectroscopy reveals a decrease in charge carrier concentration due to absorption features corresponding to neutral (only for EMIM DCA treatment) and polaronic states (for both EMIM DCA and EMIM BF₄ treatment) appearing in the visible range. As opposed to this, treatment with EMIM TCB only shows changes in electrical conductivities, while no significant improvement of the Seebeck coefficient is detected. The latter finding is in line with the UV-Vis spectra not showing additional features upon EMIM TCB treatment. Changes in the Seebeck coefficients are qualitatively linked to the structure of the anion, with regard to their charge localization and size, with stronger localization of the negative charge indicating stronger interactions with PEDOT, in addition to the basicity of the anions also having an influence.

GIWAXS measurements are performed regarding changes in average π - π -stacking distances of PEDOT chains, PEDOT-to-PSS ratio and edge-on-to-face-on ratio, since all of these quantities influence the films' ability to transport charges. The GIWAXS analysis confirms the increase in electrical conductivities as result from changes in the structure, with the trends for the different investigated quantities varying with the type of IL. π - π -stacking distances, especially in edge-on

direction are smallest for samples treated with EMIM TCB and largest for EMIM BF₄ treatment. All investigated ILs show PSS removing capabilities, thereby also contributing to the improved electrical conductivities as function of IL concentration. PEDOT chains are reoriented upon treatment with EMIM DCA and EMIM TCB, and to lesser extent also for EMIM BF₄, therefore improving charge transport along the substrate and explaining the improvement of electrical conductivities in all cases.

The influence of the anion on both the Seebeck coefficient and electrical conductivity of treated PEDOT:PSS films is clearly seen. As a consequence, our study opens a new window for a rational choice of ILs for the improvement of thermoelectric properties in PEDOT:PSS thin films. An argument can be made that combinations of anions and cations can be found in the vast library of ILs, such that the individual components more efficiently interact with PEDOT:PSS in terms of simultaneous primary and secondary doping. In addition, since both PEDOT:PSS and ILs are binary systems, their interactions are expected to be complex. Kinetic studies could help in the improvement of understanding of the underlying enhancement mechanism, and details of the interactions.

6. Experimental Section

Materials: Aqueous PEDOT:PSS dispersion (PH1000, PEDOT-to-PSS ratio = 1:2.5) was purchased from Ossila Ltd. (UK), while tetrahydrofuran ($\geq 99.9\%$, THF) was purchased from Sigma-Aldrich, along with the ionic liquids (ILs) 1-ethyl-3-methylimidazolium (EMIM) dicyanamide (DCA, $\geq 98.5\%$) and tetrafluoroborate ($\geq 97.0\%$, BF₄). 1-ethyl-3-methylimidazolium tetracyanoborate (TCB) was purchased from Merck. Thermoelectric thin films were prepared on clean soda-lime glass substrates (Carl Roth GmbH + Co. KG).

Sample preparation: PEDOT:PSS were prepared by addition of the fluorosurfactant Zonyl, as reported previously.^{46,46} PEDOT:PSS thin films were fabricated by spin-casting on clean soda-lime glass substrates at 1500 rpm for 60 s, followed by an annealing step at 140 °C for 10 minutes. This step was repeated in order to obtain thicker films. Solutions of ionic liquids in THF were prepared by adding different molar amounts of the respective ILs to THF, and performing a brief ultrasonication step prior to deposition. Post-treatment was performed by drop-casting 500 μ L of ILs in THF on the pre-formed polymer thin film, letting take effect for 1 min, drop-casting another 500 μ L and then immediately spin-casting with the parameters given above. Another annealing step was performed at 140 °C for 10 min, in order to ensure removal of THF: After performing the thermoelectric characterization (Seebeck coefficient, electrical conductivity, thickness), the samples were washed with pure THF, in order to remove possible remnants of ILs on the polymer films, and then to study the effect on the thermoelectric properties. This was done by drop-casting 200 μ L of pure THF, letting take effect for 1 min, drop-casting another 200 μ L and then immediately spinning off. After the last annealing step at 140 °C for 10 min, the samples were subjected to spectroscopical and morphological investigation.

Thermoelectric characterization: Seebeck measurements were performed by applying conductive silver paste to the edges of the polymer films and subjecting them to a static temperature gradient of 80 °C, as reported previously.³¹ The sheet resistance was measured by employing a four-point probe setup on six different locations of each sample and averaging the values. The thicknesses at the corresponding locations were measured using a DekTak (Bruker corporation) profilometer.

UV-Vis spectroscopy: A Lambda 35 spectrometer by PerkinElmer was used in a wavelength range of 350 nm to 1100 nm, with a scanning speed of 480 nm/min.

Grazing-incidence wide-angle x-ray scattering (GIWAXS): GIWAXS measurements of IL-treated PEDOT:PSS films were performed at the Austrian SAXS beamline of Elettra-Sincrotrone Trieste, at a beam energy of 8 keV, an incidence angle of 0.4° and sample-detector distance of 295.8 mm. The obtained two-dimensional (2D) scattering data were evaluated using the GIXSGUI software.⁵⁸ Sector integration was performed in an azimuthal range of -15° to 15° (with 0° being perpendicular to the substrate surface) for investigation of the structure in face-on direction and in an azimuthal range of 70° to 80° for investigation of structure in edge-on direction. PEDOT-to-PSS ratios and edge-on-to-face-on ratios are obtained from curve fitting with two Gaussian curves and a function accounting for the substrate contribution in a Levenberg-Marquardt procedure.

ASSOCIATED CONTENT

Supporting Information. The Supporting Information is available electronically free of charge. It contains a list of physical properties of the employed ionic liquids (ILs) and extracted scattering curves from the 2D GIWAXS patterns in vertical and horizontal direction.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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